MULTI-SITE SOLAR PV PORTFOLIO (35.0 kW)

Date: 22-Sep-2026

Portfolio: Integrated Portfolio Tech (2 Sites, 3 Buildings) | Location: Karnataka | Grid-Tied LT Dossier

PLANT CAPACITY	ANNUAL YIELD	CAPITAL COST	PAYBACK PERIOD
400W Mono	Specific Yield	12-Item BOM	25Y Lifecycle
35.04 kW	**55,557 kWh**	**Rs. 1,604,975**	**4Y 7M**
31.20 kWp Installed	CUF 18.1% | PR 75%	Rs. 16.05 Lakhs	Profit: Rs. 163.67L

[SITE] Customer & Site Engineering Parameters

Verified project configuration and sizing pathway

Company / Customer Name	Integrated Portfolio Tech
Location / Address	Karnataka
Contact Email Address	Not Specified
Design Path Selected	Rooftop Area Analysis
System Target Capacity	35.04 kW (2 Sites Portfolio)
Total Installed Array Rating	31.20 kWp
Solar PV Module Rating	400W Mono PERC / TOPCon Half-Cut
Total Module Count	78 Nos.

[METRICS] System Performance & Economic Evaluation

25-Year performance metrics and grid benchmark comparisons

Total Rooftop Area	4525.00 Sq.Feet (420.39 sq.m)
Utilized Usable Rooftop Area	3847 Sq.Feet (357.40 sq.m)
First Year Annual Generation	55,557.32 kWh
25-Year Cumulative Generation	1,234,364.97 kWh
Estimated Payback Period	4 Years & 7 Months
Post-Payback Profitability (25 Yrs)	Rs. 16,366,798.69 (INR 163.67 Lakhs)
Alternative FD 6.5% Benchmark (25 Yrs)	Rs. 7,748,337.97 (INR 77.48 Lakhs)

ENVIRONMENTAL & SUSTAINABILITY IMPACT (25-YEAR LIFECYCLE)

Carbon Offset: 1012 Tonnes CO2 | Tree Equivalence: 1619 Mature Trees | Green Energy: 100% Clean LT Power

Based on standard Central Electricity Authority (CEA) grid emission and forestry carbon sequestration benchmarks.

[PORTFOLIO] Multi-Site Portfolio Allocation & Regulatory Schedule

Aggregated LT solar portfolio overview, site-wise capacities, and statutory compliance

TOTAL SITES	TOTAL CAPACITY	TURNKEY CAPITAL	ANNUAL YIELD
PORTFOLIO	AGGREGATE	12-ITEM BOM	CLEAN POWER
2 Sites	**35.04 kW**	**Rs. 1,604,975**	**55,557 kWh**
3 Buildings Total	78 Panels (400W)	Rs. 16.05 Lakhs	1st Year Aggregate

[ALLOCATION] Site-by-Site Allocation & Engineering Capacity Matrix

Independent LT net-meter installations conforming to 20 kW per-connection statutory ceiling

Sl.	Site Name & Location	Bldgs	Gross Area	Capacity	Inverter	1st Yr Yield	Turnkey Cost
01	Site 1 - College (Shivamog..	2 Nos.	2,775 sq.ft	20.00 kW *	14 kW 3-Ph	31,711 kWh	Rs. 948,892
02	Site 2 - Hostel (Bengaluru)	1 Nos.	1,750 sq.ft	15.04 kW	11 kW 3-Ph	23,847 kWh	Rs. 656,083
	TOTAL PORTFOLIO SUMMARY		**4,525 sq.ft**	**35.04 kW**	**25 kW Total**	**55,557 kWh**	**Rs. 1,604,975**

** Marked installation(s) capped at statutory 20.0 kW Low Tension (LT) ceiling in accordance with KERC regulations.*

KERC & ESCOM LOW TENSION (LT) 20 kW REGULATORY COMPLIANCE MANDATE

Under Karnataka Electricity Regulatory Commission (KERC) Rooftop Solar PV regulations and ESCOM interconnect guidelines, individual Low Tension (LT) net-metered service connections are subject to a statutory capacity ceiling of 20.0 kW per consumer installation / meter account. Interconnections above 20 kW necessitate High Tension (HT 11 kV) infrastructure, dedicated transformer bays, and elevated statutory approvals.

To maintain full regulatory compliance while meeting multi-site commercial / institutional power demand, this portfolio is architected into discrete, independent LT installations. Each individual site operates with its own dedicated bidirectional net-meter, solar inverter, ACDB protection board, and safety switchgear, while engineering specifications, equipment warranties, and turnkey financials are consolidated within this unified dossier.

LT PORTFOLIO STATUTORY & GRID INTERCONNECTION STANDARDS

Metering Architecture:	Independent Bi-Directional TOD Net-Meters per site (1 - 20 kW LT Class)
Anti-Islanding Protection:	Mandatory automatic grid disconnection upon utility outage (IEC 62116 / IEEE 1547 compliant)
Switchgear & Isolation:	Dedicated IP65 ACDB with Type-II AC/DC SPDs & 10 kA Breaking MCBs per connection
DISCOM Permitting:	Separate online DISCOM portal filings with aggregated turnkey EPC execution and common monitoring

[ARRAY] Site 1 - College -- Building 1: Layout & Array Sizing

Site: Site 1 - College (Shivamogga) | Sub-Array: 13.44 kW | Site Inverter: 14 kW

Building / Array Identification	Site 1 - College -- Admin Block (Rooftop Array)
Gross Rooftop Area	1575.00 Sq.Feet (146.32 sq.m)
Utilized Effective Rooftop Area	1344 Sq.Feet (124.86 sq.m)
Building Plant Target Capacity	13.44 kW
Solar PV Module Rating	400 W Mono-PERC (Mono-facial)
Number of Solar Modules	34 Nos.
String Sizing Architecture	18 - 20 Panels per MPPT String (Optimal Voltage Band)
Selected Site Inverter Rating	14 kW, 415V 3-Ph MPPT

[INVERTER] Selected Grid-Tied Inverter Specifications (14 kW)

Core inverter rating (70% derated selection), MPPT topology & models

Selected Inverter Rating	14 kW Grid-Tied Inverter (Three-Phase)
Certified Top-Tier OEM Models	Deye, Delta Solar
Derated AC Design Power (70% Factor)	14.00 kW AC (from 13.44 kWp DC)
AC Output Voltage & Grid Phase	415V, 3-Phase, 50 Hz AC
MPPT Operating Topology	Dual MPPT Architecture (>99.5% MPPT Tracking Efficiency)
Enclosure & Grid Protection	IP65 Weatherproof Enclosure with Anti-Islanding (IEC 62116)
Inverter to ACDB Cable	3.5Cx16 Sq.mm
DC Array Isolator Switch	63A, MCB Type, 4P 415V

[MCB] Selected MCB & Switchgear Protection Specifications

Breaker sizing, trip characteristics, and surge protection devices

Incomer Overcurrent Protection	Thermal-Magnetic MCB / MCCB with Manual Isolation
Breaker Current Rating	25 Amps
Rated Inverter Operating Current	19.48 Amps
Trip Curve & Breaking Capacity	Type C Curve / 10 kA Breaking Capacity (IS/IEC 60898-1)
Surge Protection Device (SPD)	Type II AC SPD (40 kA, 415V / 230V) with Remote Signalling

[CABLES] Electrical Cable Sizing & Interconnection Schedule

Cable conductor specifications, insulation, and voltage drop compliance

Solar PV DC Array Cable	4 / 6 sq.mm UV-Resistant Solar DC Cable (1.5 kV, EN 50618)
Inverter AC Output Cable	3.5Cx16 Sq.mm (Cu / XLPE / Armoured)
ACDB to Utility Grid Cable	3.5Cx6 Sq.mm (Al / XLPE / Armoured)
Earth Grounding Conductor	16 sq.mm Cu / GI Earth Conductor connected to Dual Chemical Pits
Design Standard & Voltage Drop	Voltage Drop < 2% in accordance with IS 694 / IEC 60364-5-52

[ACDB] AC Distribution Board (ACDB 1 IN / 1 OUT) Specifications

Switchgear enclosure, busbar rating, and metering point

ACDB Enclosure Specification	IP65 Weatherproof Polycarbonate / Sheet Steel
Internal Busbar Rating	50A, 415V, 3PN Aluminium
Incomer & Outgoing Terminations	Cable Glands with Compression Lugs & Dual Earthing
Interconnection Distance	25.0 Meters (ACDB to Utility Net-Meter)

[ARRAY] Site 1 - College -- Building 2: Layout & Array Sizing

Site: Site 1 - College (Shivamogga) | Sub-Array: 9.99 kW | Site Inverter: 14 kW

Building / Array Identification	Site 1 - College -- Lab Block (Rooftop Array)
Gross Rooftop Area	1200.00 Sq.Feet (111.48 sq.m)
Utilized Effective Rooftop Area	999 Sq.Feet (92.81 sq.m)
Building Plant Target Capacity	9.99 kW
Solar PV Module Rating	580 W N-Type TOPCon
Number of Solar Modules	18 Nos.
String Sizing Architecture	18 - 20 Panels per MPPT String (Optimal Voltage Band)
Selected Site Inverter Rating	14 kW, 415V 3-Ph MPPT

[INVERTER] Selected Grid-Tied Inverter Specifications (14 kW)

Core inverter rating (70% derated selection), MPPT topology & models

Selected Inverter Rating	14 kW Grid-Tied Inverter (Three-Phase)
Certified Top-Tier OEM Models	Deye, Delta Solar
Derated AC Design Power (70% Factor)	14.00 kW AC (from 9.99 kWp DC)
AC Output Voltage & Grid Phase	415V, 3-Phase, 50 Hz AC
MPPT Operating Topology	Dual MPPT Architecture (>99.5% MPPT Tracking Efficiency)
Enclosure & Grid Protection	IP65 Weatherproof Enclosure with Anti-Islanding (IEC 62116)
Inverter to ACDB Cable	3.5Cx16 Sq.mm
DC Array Isolator Switch	63A, MCB Type, 4P 415V

[MCB] Selected MCB & Switchgear Protection Specifications

Breaker sizing, trip characteristics, and surge protection devices

Incomer Overcurrent Protection	Thermal-Magnetic MCB / MCCB with Manual Isolation
Breaker Current Rating	25 Amps
Rated Inverter Operating Current	19.48 Amps
Trip Curve & Breaking Capacity	Type C Curve / 10 kA Breaking Capacity (IS/IEC 60898-1)
Surge Protection Device (SPD)	Type II AC SPD (40 kA, 415V / 230V) with Remote Signalling

[CABLES] Electrical Cable Sizing & Interconnection Schedule

Cable conductor specifications, insulation, and voltage drop compliance

Solar PV DC Array Cable	4 / 6 sq.mm UV-Resistant Solar DC Cable (1.5 kV, EN 50618)
Inverter AC Output Cable	3.5Cx16 Sq.mm (Cu / XLPE / Armoured)
ACDB to Utility Grid Cable	3.5Cx6 Sq.mm (Al / XLPE / Armoured)
Earth Grounding Conductor	16 sq.mm Cu / GI Earth Conductor connected to Dual Chemical Pits
Design Standard & Voltage Drop	Voltage Drop < 2% in accordance with IS 694 / IEC 60364-5-52

[ACDB] AC Distribution Board (ACDB 1 IN / 1 OUT) Specifications

Switchgear enclosure, busbar rating, and metering point

ACDB Enclosure Specification	IP65 Weatherproof Polycarbonate / Sheet Steel
Internal Busbar Rating	50A, 415V, 3PN Aluminium
Incomer & Outgoing Terminations	Cable Glands with Compression Lugs & Dual Earthing
Interconnection Distance	25.0 Meters (ACDB to Utility Net-Meter)

[ARRAY] Site 2 - Hostel -- Building 1: Layout & Array Sizing

Site: Site 2 - Hostel (Bengaluru) | Sub-Array: 15.04 kW | Site Inverter: 11 kW

Building / Array Identification	Site 2 - Hostel -- Hostel A (Rooftop Array)
Gross Rooftop Area	1750.00 Sq.Feet (162.58 sq.m)
Utilized Effective Rooftop Area	1504 Sq.Feet (139.73 sq.m)
Building Plant Target Capacity	15.04 kW
Solar PV Module Rating	550 W Bifacial (Mono / TOPCon)
Number of Solar Modules	28 Nos.
String Sizing Architecture	18 - 20 Panels per MPPT String (Optimal Voltage Band)
Selected Site Inverter Rating	11 kW, 415V 3-Ph MPPT

[INVERTER] Selected Grid-Tied Inverter Specifications (11 kW)

Core inverter rating (70% derated selection), MPPT topology & models

Selected Inverter Rating	11 kW Grid-Tied Inverter (Three-Phase)
Certified Top-Tier OEM Models	Deye 11K-G04, Kstar Solar
Derated AC Design Power (70% Factor)	10.53 kW AC (from 15.04 kWp DC)
AC Output Voltage & Grid Phase	415V, 3-Phase, 50 Hz AC
MPPT Operating Topology	Dual MPPT Architecture (>99.5% MPPT Tracking Efficiency)
Enclosure & Grid Protection	IP65 Weatherproof Enclosure with Anti-Islanding (IEC 62116)
Inverter to ACDB Cable	3.5Cx16 Sq.mm
DC Array Isolator Switch	50A, MCB Type, 4P 415V

[MCB] Selected MCB & Switchgear Protection Specifications

Breaker sizing, trip characteristics, and surge protection devices

Incomer Overcurrent Protection	Thermal-Magnetic MCB / MCCB with Manual Isolation
Breaker Current Rating	25 Amps
Rated Inverter Operating Current	14.65 Amps
Trip Curve & Breaking Capacity	Type C Curve / 10 kA Breaking Capacity (IS/IEC 60898-1)
Surge Protection Device (SPD)	Type II AC SPD (40 kA, 415V / 230V) with Remote Signalling

[CABLES] Electrical Cable Sizing & Interconnection Schedule

Cable conductor specifications, insulation, and voltage drop compliance

Solar PV DC Array Cable	4 / 6 sq.mm UV-Resistant Solar DC Cable (1.5 kV, EN 50618)
Inverter AC Output Cable	3.5Cx16 Sq.mm (Cu / XLPE / Armoured)
ACDB to Utility Grid Cable	3.5Cx6 Sq.mm (Al / XLPE / Armoured)
Earth Grounding Conductor	16 sq.mm Cu / GI Earth Conductor connected to Dual Chemical Pits
Design Standard & Voltage Drop	Voltage Drop < 2% in accordance with IS 694 / IEC 60364-5-52

[ACDB] AC Distribution Board (ACDB 1 IN / 1 OUT) Specifications

Switchgear enclosure, busbar rating, and metering point

ACDB Enclosure Specification	IP65 Weatherproof Polycarbonate / Sheet Steel
Internal Busbar Rating	50A, 415V, 3PN Aluminium
Incomer & Outgoing Terminations	Cable Glands with Compression Lugs & Dual Earthing
Interconnection Distance	15.0 Meters (ACDB to Utility Net-Meter)

[SITE BOM] Site 1: Site 1 - College -- Turnkey Bill of Materials (BOM)

Complete 12-item engineering specification and cost breakdown for Site 1 - College (20 kW)

*** Statutory 20 kW LT Ceiling Applied: Rooftop raw potential was 23.43 kW. Capped at 20.0 kW under KERC regulations.**

Sl.	Component / Material Description	Engineering Specification	Qty	Amount (INR)
01	Solar PV modules	Admin Block: 400W Mono-PERC (Mono-facial) (34 Nos.) + Lab Block: 580W N-Type	52 Nos.	Rs. 437,092
02	On-grid inverter	14 kW three-phase (Admin Block: 10kW + Lab Block: 8kW)	2 Nos.	Rs. 147,000
03	Rooftop mounting structure	HDGI, module-compatible	2 Roof Set(s)	Rs. 95,000
04	DC solar cable	6 sq.mm copper, PV-rated	220 m	Rs. 18,700
05	AC cable	35 sq.mm copper	110 m	Rs. 49,500
06	MC4 connectors	PV-rated connector pairs	52 pairs	Rs. 15,600
07	DCDB	Isolator + fuse + Type-II DC SPD	2 No(s).	Rs. 18,000
08	ACDB	Admin Block: 25A MCB + Lab Block: 25A MCB + Type-II AC SPD	2 Set(s)	Rs. 18,000
09	Earthing kit	Earth electrodes & accessories (IS 3043 - 3 Pits)	3 Nos.	Rs. 9,000
10	Lightning arrester	Standard rooftop PV type	1 No.	Rs. 8,000
11	Installation accessories	Lugs, ferrules, clamps, conduit, glands, etc.	1 Lot	Rs. 38,000
12	Installation & commissioning	Labour and commissioning	1 Job	Rs. 95,000
			TOTAL PLANT-SIDE CAPITAL COST (Site 1 - College)	**Rs. 948,892.00**

[UTILITY] ESCOM / Utility Statutory Approvals & Grid Budget (Site 1 - College)

Official DISCOM statutory filing fees and net-metering charges for this connection

Application & Registration Fee	Rs. 1,000 + GST (DISCOM portal filing)
Facilitation & Technical Processing Fee	Rs. 2,000 + GST (Feasibility clearance)
PPA e-Sign Stamp Duty Charge	Rs. 11.8 (Electronic stamping)
Bidirectional Utility Net-Meter	Rs. 12,000 (DISCOM meter testing/procurement)
Refundable Security Deposit	Rs. 8,000 (Refundable utility escrow)

TOTAL ESTIMATED CAPITAL OUTLAY FOR SITE 1 - COLLEGE

Turnkey Plant BOM: Rs. 948,892 | ESCOM Budget: Rs. 23,012 | Site Net Total: Rs. 971,904 (Rs. 9.72 Lakhs)

[SITE BOM] Site 2: Site 2 - Hostel -- Turnkey Bill of Materials (BOM)

Complete 12-item engineering specification and cost breakdown for Site 2 - Hostel (15.04 kW)

Sl.	Component / Material Description	Engineering Specification	Qty	Amount (INR)
01	Solar PV modules	550W Bifacial (Mono / TOPCon)	28 Nos.	Rs. 280,000
02	On-grid inverter	11 kW, three-phase (Deye 11K-G04 or equiv.)	1 No.	Rs. 117,333
03	Rooftop mounting structure	HDGI, module-compatible	1 Set	Rs. 70,000
04	DC solar cable	6 sq.mm copper, PV-rated	160 m	Rs. 13,600
05	AC cable	25 sq.mm copper	85 m	Rs. 29,750
06	MC4 connectors	PV-rated connector pairs	28 pairs	Rs. 8,400
07	DCDB	Isolator + fuse + Type-II DC SPD	1 No.	Rs. 14,000
08	ACDB	25A MCB + 40A RCCB + Type-II AC SPD	1 No.	Rs. 14,500
09	Earthing kit	Earth electrodes & accessories (IS 3043 - 3 Pits)	3 Nos.	Rs. 7,500
10	Lightning arrester	Standard rooftop PV type	1 No.	Rs. 6,000
11	Installation accessories	Lugs, ferrules, clamps, conduit, glands, etc.	1 Lot	Rs. 25,000
12	Installation & commissioning	Labour and commissioning	1 Job	Rs. 70,000
		TOTAL PLANT-SIDE CAPITAL COST (Site 2 - Hostel)		**Rs. 656,083.33**

[UTILITY] ESCOM / Utility Statutory Approvals & Grid Budget (Site 2 - Hostel)

Official DISCOM statutory filing fees and net-metering charges for this connection

Application & Registration Fee	Rs. 1,000 + GST (DISCOM portal filing)
Facilitation & Technical Processing Fee	Rs. 2,000 + GST (Feasibility clearance)
PPA e-Sign Stamp Duty Charge	Rs. 11.8 (Electronic stamping)
Bidirectional Utility Net-Meter	Rs. 12,000 (DISCOM meter testing/procurement)
Refundable Security Deposit	Rs. 8,000 (Refundable utility escrow)

TOTAL ESTIMATED CAPITAL OUTLAY FOR SITE 2 - HOSTEL

Turnkey Plant BOM: Rs. 656,083 | ESCOM Budget: Rs. 23,012 | Site Net Total: Rs. 679,095 (Rs. 6.79 Lakhs)

[PORTFOLIO TOTAL] Consolidated Portfolio Capital Cost & Grand Total Investment

Consolidated turnkey capital outlay across all portfolio sites and DISCOM budgets

TOTAL CAPACITY	TURNKEY BOM	ESCOM BUDGET	TOTAL CAPITAL
AGGREGATE	12-ITEM BOM	UTILITY	GRAND TOTAL
35.04 kW	**Rs. 16.05L**	**Rs. 0.46L**	**Rs. 16.51L**
2 Sites Sized	Rs. 1,604,975	Rs. 46,024	Rs. 1,650,999

[CAPEX MATRIX] Portfolio Site-by-Site Capital Investment Matrix

Comprehensive turnkey capital expenditure required for all sites

Sl.	Site Name & Location	Buildings	Capacity	Plant BOM Cost	ESCOM Budget	Total Outlay
01	Site 1 - College	2 Nos.	20.00 kW *	Rs. 948,892	Rs. 23,012	Rs. 971,904
02	Site 2 - Hostel	1 Nos.	15.04 kW	Rs. 656,083	Rs. 23,012	Rs. 679,095
GRAND TOTAL CAPITAL REQUIRED (ALL SITES)			**35.04 kW**	**Rs. 1,604,975**	**Rs. 46,024**	**Rs. 1,650,999**

** Marked installation(s) capped at statutory 20.0 kW Low Tension (LT) ceiling in accordance with KERC regulations.*

PORTFOLIO PROCUREMENT, EPC EXECUTION & DISCOM GRID CONNECTIVITY

All electrical installations conform to CEA Technical Standards for Distributed Generation and State Grid Code. Turnkey procurement bundles Tier-1 TOPCon modules and high-efficiency inverters under consolidated bulk pricing, while each site operates under its dedicated independent DISCOM net-meter account (RR number). This avoids High Tension (HT 11 kV) transformer infrastructure and maintains complete statutory compliance.

[FINANCE] 25-Year Financial Amortization Schedule

Annual cash flow model incorporating 1%/year degradation & tariff escalation

CAPITAL INVESTMENT (CAPEX)	PAYBACK PERIOD (RETURN)	25Y NET SAVINGS (PROFIT)	FD 6.5% BENCHMARK (BENCHMARK)
Rs. 1,604,975	**4Y 7M**	**Rs. 163.67L**	**Rs. 77.48L**
12-Item BOM Basis	Break-Even Year	Rs. 16,366,799	Alternative Asset

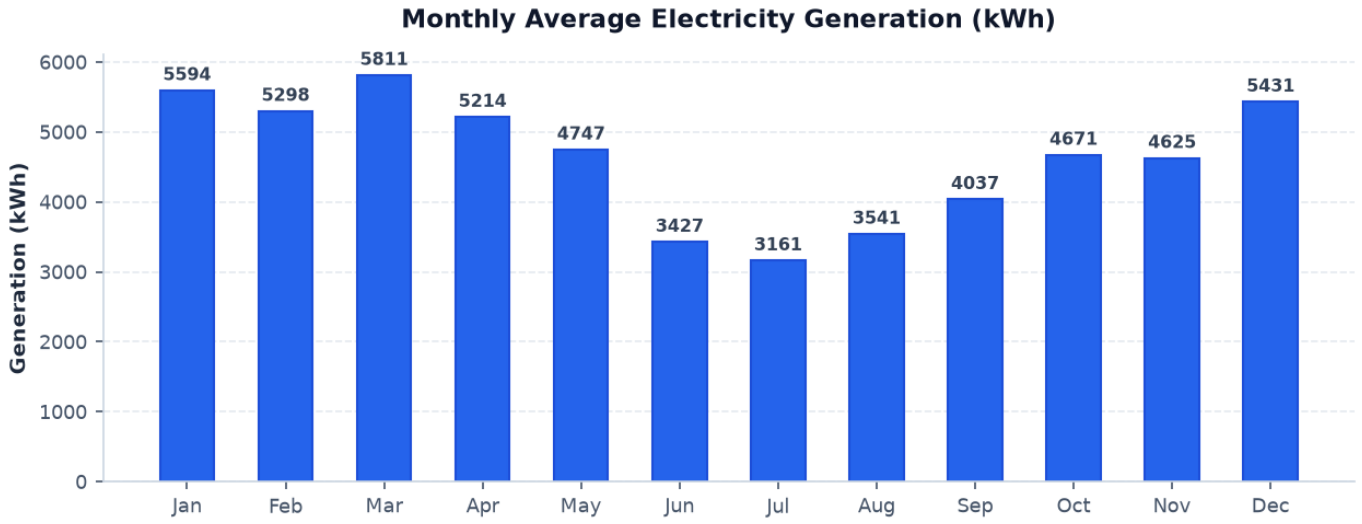
Yr	Gen (kWh)	Tariff (Rs/u)	Gross Rev (Rs)	AMC (Rs)	Net Annual (Rs)	Cumulative Net (Rs)
01	55,557	Rs. 8.05	447,236	21,024	426,212	426,212
02	55,002	Rs. 8.45	464,765	21,024	443,741	869,953
03	54,452	Rs. 8.87	482,987	21,024	461,963	1,331,916
04 *	**53,907**	**Rs. 9.32**	**502,415**	**21,024**	**481,391**	**1,813,307**
05	53,368	Rs. 9.78	521,940	21,024	500,916	2,314,224
06	52,834	Rs. 10.27	542,610	21,024	521,586	2,835,810
07	52,306	Rs. 10.78	563,860	21,024	542,836	3,378,645
08	51,783	Rs. 11.32	586,184	21,024	565,160	3,943,806
09	51,265	Rs. 11.89	609,543	21,024	588,519	4,532,325
10	50,753	Rs. 12.48	633,392	21,024	612,368	5,144,693
11	50,245	Rs. 13.11	658,713	21,024	637,689	5,782,382
12	49,743	Rs. 13.76	684,458	21,024	663,434	6,445,816
13	49,245	Rs. 14.45	711,593	21,024	690,569	7,136,384
14	48,753	Rs. 15.18	740,066	21,024	719,042	7,855,427
15	48,265	Rs. 15.93	768,864	21,024	747,840	8,603,267
16	47,783	Rs. 16.74	799,880	21,024	778,856	9,382,123
17	47,305	Rs. 17.57	831,144	21,024	810,120	10,192,243
18	46,832	Rs. 18.45	864,044	21,024	843,020	11,035,263
19	46,363	Rs. 19.37	898,058	21,024	877,034	11,912,297
20	45,900	Rs. 20.34	933,600	21,024	912,576	12,824,873
21	45,441	Rs. 21.36	970,614	21,024	949,590	13,774,463
22	44,986	Rs. 22.43	1,009,043	21,024	988,019	14,762,482
23	44,536	Rs. 23.55	1,048,833	21,024	1,027,809	15,790,291
24	44,091	Rs. 24.73	1,090,372	21,024	1,069,348	16,859,640
25	43,650	Rs. 25.96	1,133,158	21,024	1,112,134	17,971,774

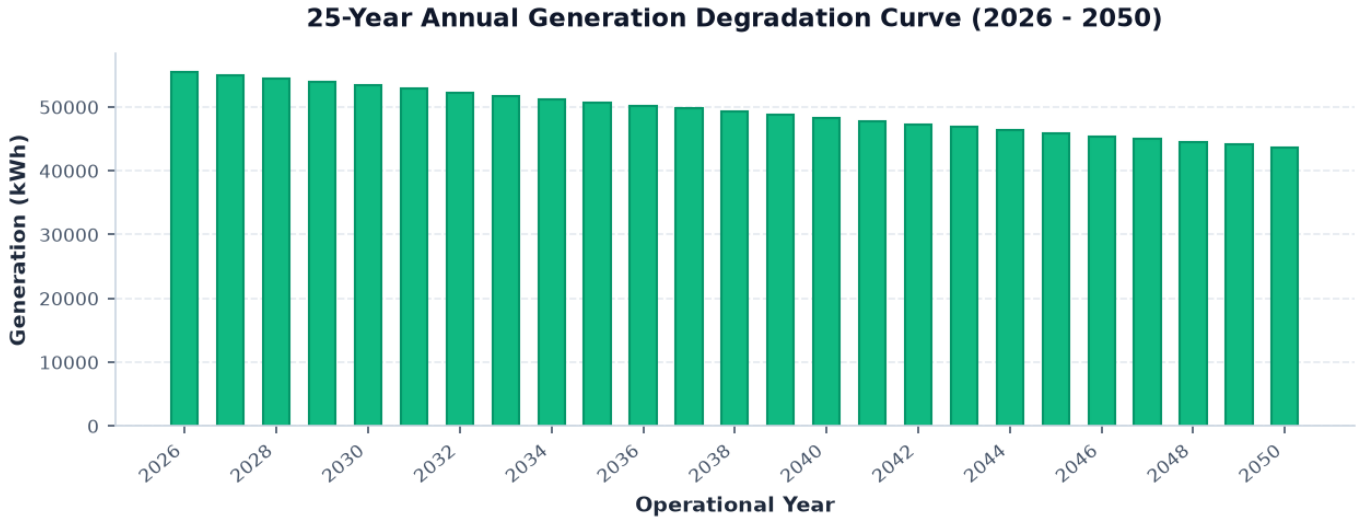
** Highlighted row indicates capital payback year. Annual maintenance contract (AMC) budgeted at Rs. 600/kW/year with 1% annual PV degradation.*

[ANALYTICS] Generation Analytics & Performance Curves

Monthly solar insolation yield and 25-year degradation profile

SPECIFIC YIELD (YIELD)	PERFORMANCE RATIO (PR)	CAPACITY UTIL. (CUF)	DEGRADATION RATE (DEGRADATION)
1586 kWh/kWp	**75.0%**	**18.1%**	**1.0% / year**
kWh/kWp/Year	IEC 61724 Standard	Annual Plant Factor	Tier 1 Linear Warranty





Simulated with regional GHI solar insolation database (3.26 to 5.40 kWh/m2/day). System performance calculated using PVsyst/MNRE benchmarks.

VoltRoof Designs
CLEAN ENERGY. BRIGHTER TOMORROW

ON-GRID SOLAR PV SYSTEM

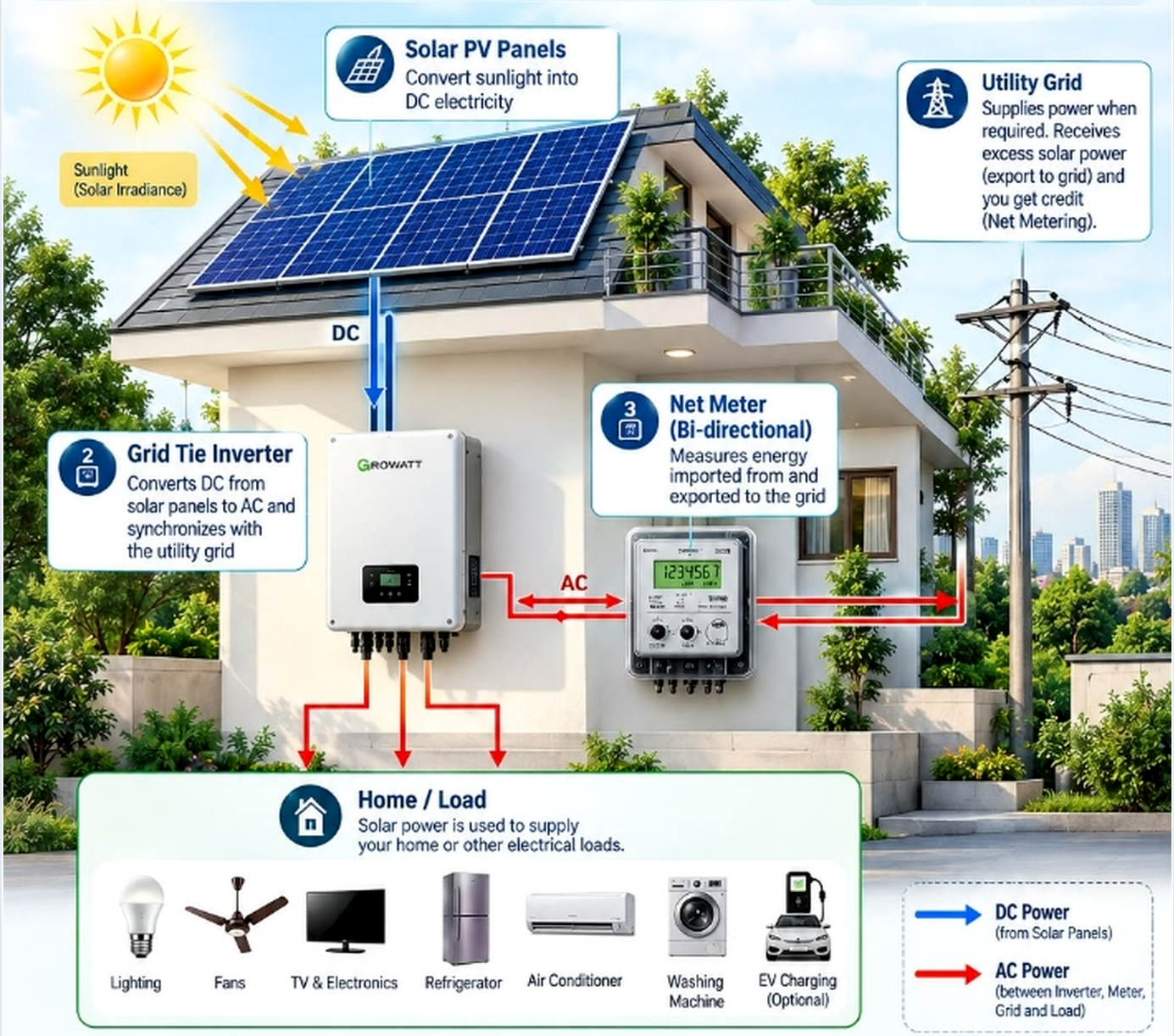
(Grid Connected Solar System)

GENERATE • CONSUME • EXPORT • SAVE

Harness The Sun For A Better Tomorrow

An on-grid solar PV system is connected to the utility grid. The solar power generated is used to meet your load demand. Any excess power is exported to the grid and you get credit through net metering. No battery is used in this system.

- Lower Electricity Bills
- Clean Energy
- Sustainable Future



KEY ADVANTAGES

- Reduced Electricity Bills
- Lower Carbon Emissions
- No Battery Required
- Net Metering Benefits
- Low Maintenance
- Environment Friendly

VoltRoof Designs

Clean Energy Better Tomorrow